INCH-POUND

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MILITARY SPECIFICATION

DISPLAY CASES, MECHANICALLY REFRIGERATED, SELF-SERVICE

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-D-40605C. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers refrigerated, self-service, food display cases (see 6.1).

1.2 Classification. Display cases covered by this specification shall be of the following types, styles, classes, superstructures (non-refrigerated), suffixes, and sizes, as specified (see 6.2). The combination of available styles, classes, superstructures, and suffixes are covered herein:

Types

Type I - Produce
Type II - Dairy Products
Type III - Meat
Type IV - Frozen Food or Meat
Type V - Ice Cream

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Styles

Style A - Single deck - Wall installation
Class 1 - Without additional refrigerator shelves (top discharge air curtain)
Class 2 - One additional refrigerator shelf (top discharge air curtain)
Class 3 - Two additional refrigerator shelves (top discharge air curtain)
Class 4 - Three additional refrigerator shelves (top discharge air curtain)
Class 5 - Four additional refrigerator shelves (top discharge air curtain)
Class 6 - Five additional refrigerator shelves (top discharge air curtain)

Style B - Narrow island, one deck, one case

Style C - Double wide, island case

Style D - Wide island, back-to-back cases

Style E - Multi deck, dairy-deli, end display
Class 4 - Three adjustable shelves

Style F - Single deck, air curtain cooler
Class 4 - Three additional refrigerator shelves

Style G - Bottomless cooler with roll-in carts

Style H - End case, for types IV and V, styles C and D

Superstructure (Sup) (Not refrigerated)

Sup M-O - No superstructure required
Sup M-1 - High canopy, fixed slanted mirror (available only for style A, classes 1 and 2 in all types except II)
Sup M-2 - Low canopy, fixed, slanted mirror (available only for style A, class 1 in all types except II)
Sup M-3 - With sliding, slanted mirror (available only for style A, class 1 in all types, except II)
Sup M-4 - With sliding, vertical, clear glass (available only for style A, class 1 in all types, except II)
Sup M-5 - High canopy, with shelves (available for style A, class 1 in all types, except II)
Sup M-6 - Low canopy, with shelves (available only for style A, class 1 in all types except II)
Sup M-7 - Truss type shelves (available for type IV and V in style C and D only)

Suffix SF - Storage below deck - Front access door(s) (available for type I and III style A only)

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Suffix SR - Storage below deck - Rear access door(s) (available for type I and III style A only)

Suffix HRLD - Vertical insulated back with insulated hinged rear loading doors (types I, II and III only)

Suffix SRLD - Vertical insulated back with noninsulated sliding doors for loading from refrigerated area (types I, II and III only)

Sizes

Size 2 - For 2 roll-in racks (type II, style G only)
Size 3 - For 3 roll-in racks (type II, style G only)
Size 5 - 5 foot long (types II, III and IV, styles B and E only)
Size 6 - 6 foot long (types II and III, style A, class 6 and types IV and V, styles C, D, and E)
Size 8 - 8 foot long (all types and styles)
Size 12 - 12 foot long (all types and styles)

Ambient temperature

Air-conditioned area - 78°F and 55 percent relative humidity
Non air-conditioned area - 90°F

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
TT-P-664	-	Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC- Compliant
MMM-A-260	-	Adhesive, Water-Resistant, (for Sealing Waterproofed Paper)
PPP-B-621	-	Boxes, Wood: Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible

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MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Methods of
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaperproof or Waterproofed, Flexible
MIL-R-43979	-	Racks, Milk Container, Mobile and Racks, Egg Container, Mobile

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 117 - Salt Spray (Fog) Testing
D 3951 - Standard Practice for Commercial Packing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 471 - Commercial Refrigerators and Freezers

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062.)

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COMMERCIAL REFRIGERATION MANUFACTURERS ASSOCIATION (CRMA)

Recommended Guidelines for Retail Food Store Design

Voluntary Minimum Standard for Retail Food Store Refrigerator, Health and Sanitation CRS-S1-86

(Applications for copies should be addressed to the National Technical Committee, 1101 Connecticut Ave, N.W. Washington, DC 20036.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard product. The display cases covered by this specification shall be the current standard product of the manufacturer except for any design modifications specified herein.

3.2 Codes and standards. The display cases, including display cases with open type compressors, and display cases with remote condensing units, shall comply with the applicable requirements of Underwriters Laboratories (UL) Standard No. 471 (see 4.3) and with the National Fire Protection Association (NFPA) Standard No. 70 (see 3.4.20).

3.3 Materials and components. The materials and components used shall conform to the applicable referenced specifications, standards, and requirements specified herein. Where materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice provided the completed items comply with all provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

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3.3.1 Insulation. Insulation shall be industrial type fiberglass, expanded polystyrene, or expanded polyurethane, shall be homogeneous throughout, and shall have a "K" factor rating determined by using a 75°F mean temperature and expressed in British thermal units (Btu) per hour, per square foot, per degree temperature differential, per inch of thickness. Loose fill type insulation shall not be acceptable. Flame retardant grades shall be used where needed to meet requirements of UL 471.

3.3.1.1 Rigid. Rigid insulation shall be pre-expanded polyurethane or pre-expanded polystyrene having a manufacturer's commercial rated "K" factor of not more than 0.15 for polyurethane and 0.25 for polystyrene and a minimum core density of 1.8 pounds per cubic foot (see 4.4.6).

3.3.1.2 Foamed-in-place. Foamed-in-place insulation shall be polyurethane foam having a manufacturer's commercial rated "K" factor of not more than 0.15. The foam shall have a density of not less than 1.7 pounds per cubic foot when frothed and 1.9 pounds per cubic foot when poured with the density tested as specified in 4.4.6.

3.3.1.3 Semirigid. Semirigid insulation shall be "B" fiber or finer industrial type fiberglass having a manufacturer's rated "K" factor of not more than 0.25 for polyurethane and 1.0 for polystyrene and a minimum core density of 3.0 pounds per cubic foot and shall be held together with a thermosetting binder.

3.3.2 Refrigerant. Refrigerant used in the refrigerating system of the display cases shall conform to BB-F-1421.

3.4 Design and construction.

3.4.1 Design. Display cases shall be designed for merchandising the specific types of products as indicated in 1.2 and the types, styles, superstructures, classes, suffixes, sizes and temperature operations specified to be furnished. When a superstructure is to be furnished as part of a display case, the display case shall be designed and fabricated to accept and support the superstructure and create an appearance of an integral unit. Shelves, racks, lighting, price tag moulding, switchable anti-condensation heaters, cart bumpers, splash guard (kick-plate), drain, air curtains, and trim shall be furnished and installed as specified for a complete installation. The display cases for free standing placement shall have readily removable insulated ends and where lineups are specified, shall be designed and furnished with one set of common ends for each lineup, partitions, and fittings to allow for matching lineup installations of the same type cases. Front glass guards for front sill shall be provided on cases in accordance with manufacturer's standard practice. Unless otherwise specified (see 6.2), all glass shall be tempered glass or safety glass.

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3.4.2 Construction. Display cases shall be constructed to provide positive refrigeration and protection of contained foods in both the display and storage areas in the specified temperature range. The exterior shall be constructed to facilitate cleaning, inspection, servicing, and maintenance. The panel joints and seams shall be sealed and made smooth. Exposed rivets and screw or bolt heads shall be of the low profile type such as brazier or modified brazier rivets or pan and oval heads, respectively. The entire interior shall be arranged and assembled to be readily cleanable throughout. Removal of pans shall give accessibility for servicing and permit cleaning of coils and fans. Night covers shall be furnished for the ice cream and frozen food cases, if so designed by the manufacturer. All lighting shall be recessed on shelving to protect lamps against breakage as specified in UL 471. In addition, the lights for ice cream and frozen food cases shall have protective shields to reduce heat generation. Lighting shall be designed so only reflected light can be seen from the front of the case, and separate switching shall be provided for light circuits. Permanently attached card mouldings to hold the standard 1-1/4 inch price tags shall be installed on front edges of all shelves and deck. Moulding to hold 3 inch price cards shall be provided when specified (see 6.2). A junction box or boxes shall be installed for lights, fans, anti-condensation heaters, and defroster circuits and shall be easily accessible for maintenance and service. Heat strips shall be installed on all display cases where required to prevent formation of condensation on exterior surfaces. Horizontal cart bumpers are required on all display cases. Sensor probes for connection to the store master temperature sensor alarm system shall be installed in the air discharge system of all display cases. Sensor probes must be compatible with temperature alarm systems. Bolts, clips, pins, locking cams, fasteners, accessories, common ends, sealing kits, or other parts required for installation shall be furnished. The display cases shall be completely ready for installation or for field assembly if shipped other than completely assembled for operation. Units to be assembled at the site shall be marked and furnished with complete drawings and instructions for assembly and installation.

3.4.3 Matching lineups. Display cases for matching lineup shall be provided with joiner kits and trim, and with one set of ends (1L and 1R) for each matching lineup furnished as specified (see 6.2). Plugs and partitions shall be furnished with the display case, canopies, and superstructures when adjoining units have different operating temperatures; when single deck units adjoin double-duty cases; when adjoining units operate on different defrost cycles; or when otherwise essential for a complete installation in satisfactory operating condition. The plugs and partitions shall be insulated or noninsulated type as required with plexiglas fillers or sections in exposed areas, and shall be furnished with the necessary fasteners and trim. Materials and finish of the plugs and partitions shall match adjoining sections in the display case and superstructures. Installation of the plugs and partitions shall be made by the contractor in the shop; and shall be shipped separately only when essential to remove items for crating and shipping or when the items cannot be installed until the cases are installed and fitted together. When plugs or partitions and trim are furnished as separate items for installation at the site, such items shall be marked and complete installation instructions and diagrams shall be enclosed with the items.

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3.4.4 Physical characteristics and performance requirements.

3.4.4.1 Dimensional requirements. The dimensions shall be as specified in tables I and II. The size 5 display case shall have a nominal length of 5 feet, the size 6 display case shall have a nominal length of 6 feet, the size 8 display case shall have a nominal length of 8 feet, and the size 12 display case shall have a nominal length of 12 feet. The size 6 display case shall be used to close off ends for styles C and D.

TABLE I. Display case dimensions, outside

Style	Class	Depth inches	Height Front inches	Height Back inches
A	1	39 TO 48	31 TO 42	37 TO 44
A	2	43 TO 45	31 TO 42	71 TO 75
A	3	39 TO 48	31 TO 42	60 TO 75
A	4	39 TO 48	27 TO 35	76 TO 85
A	5	39 TO 48	17 TO 35	76 TO 85
A	6	26 TO 34	18 TO 26	80 TO 95
B		53 TO 58	33 TO 42	33 TO 42
C		77 TO 81	33 TO 42	33 TO 42
D		82 TO 88	33 TO 42	33 TO 42
E	4	40 TO 44	25 TO 29	60 TO 69
F	4	31 TO 45	15 TO 33	80 TO 90
G		44 TO 58	12 TO 18	80 TO 95
H		36 TO 44	33 TO 42	33 TO 42

TABLE II. Superstructures

Superstructure	Top height above floor (inches)	Depth at top front to back (inches)
M-1	71 - 75	27 - 36
M-2	55 - 67	15 - 21
M-3	67 - 75	20 - 30
M-4	69 - 77	15 - 26
M-5	66 - 75	21 - 35
M-6	55 - 60	15 - 21
M-7	56 - 64	28 - 36

3.4.4.2 Internal temperature requirements. The display cases shall be designed, constructed, and equipped with refrigerating systems so that the internal display case temperatures shown in table III shall be maintained in one of the display case ambient operating temperatures of 78°F and 55 percent relative humidity or 90°F as specified (see 6.2). (Night covers on type IV and type V cases may be used to meet the requirement of operation in 90°F ambient.)

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For remote units the condensing unit shall be operated in an ambient of 100°F. When tested as specified in 4.4.1.2, display cases with either 60 or 50-hertz motors shall maintain internal temperatures within the limits specified in table III with the running time of the condensing unit not to exceed 75 percent for 78°F display case ambient temperature; and 85 percent for 90°F display case ambient temperature. During conduct of the test required by 4.4.1.2, the exterior surface of the display case shall not be more than 10°F below ambient temperature (see 4.4.2). Satisfactory operation in more than one of the ambient test conditions is not a requirement of this specification. Remote condensing units for display cases shall operate in an ambient temperature of 110°F without cutting out on motor overload, and self-contained display cases shall operate in 100°F ambient temperature without cutting out on motor overload (see 4.4.1.2).

TABLE III. Display and storage (product) temperaturae

Type		Minimum	Maximum
I	Produce	32°F	45°F
II	Dairy	31°F	40°F
III	Chilled Meat	28°F	38°F
IV	Frozen Food or Meat	NONE	0°F
V	Ice Cream	NONE	-15°F

3.4.4.3 Operation. Display cases shall be operational when tested as specified in 4.4.4. The refrigerant systems shall not leak, electrical equipment shall be connected without shorts or grounds, all equipment shall be adjusted as required; and all controls shall function.

3.4.5 Display cases styles.

3.4.5.1 Double-duty cases. Display cases designated by SF or SR for double-duty shall have provision for refrigerated storage in the bottom of the case. Doors shall be provided for access to the storage space. Access doors shall be in the front or in the rear as specified (see 6.2) and may be hinged or sliding, except that aisle doors on front of cases shall be sliding type. Doors shall be insulated. Sliding type doors shall operate on stainless steel, rigid extruded plastic, or extruded aluminum tracks. Doors shall slide easily and shall be readily removable. Hinged doors shall have adjustable, chromium plated, brass hardware. The bottom storage area shall be removable, sectional, full length center shelf and floor racks. Shelves shall be supported on guides to permit shelves to be inserted or removed easily. Guides shall hold the shelves in alignment so that shelves are not displaced or allowed to fall out of position. Floor racks shall be formed and supported to allow for air circulation beneath the stored products. Door liner and outer shell shall be finished and constructed to match the inner liner and outer shell of the display case.

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3.4.5.2 Style A, class 1. Display cases shall be a single deep well deck for maximum depth display and shall have adjustable racks for at least three positions to permit shallow display of various depths. Racks for types I and III shall be adjustable for horizontal or slanted display. Wall installations shall have low fronts for maximum display.

3.4.5.2.1 Style A, class 2. The class 2 display cases shall be designed with a lower deck and one additional shelf within the refrigerated area and for wall installation, and shall have a top discharge air curtain system. The top of the case shall be insulated and designed as a shelf for display of nonrefrigerated items. The ends of the case shall be equipped with nominal 1/4 inch tempered or safety glass screen wings where required to maintain refrigerated temperatures in the case. Shelves within the refrigerated area shall be adjustable up and down or shall be designed for setting horizontally in a fixed position or for angling up or down from front to back. The class 2 display case may be specified for all types, except type IV and V.

3.4.5.2.1.1 Style A, class 1 or 2. For type 1, style A, classes 1 or 2, the following applies: A fixed slanted mirror shall be mounted the full length of the case at an angle to give full reflection of the display without reflecting any section of the floor from a straight ahead view point. Unless otherwise specified, the ends of the case shall be equipped with nominal 1/4 inch tempered glass screen wings where required to maintain refrigerated temperatures in the case. Fluorescent light fixtures, with reflectors and bulbs, shall extend the full length of both the top canopy, and the front bottom edge of the upper shelf. A separate switch shall be provided for light circuits. One produce divider shall be furnished for every 2 linear feet of display; one plastic bag roller shall be furnished for each display case; wire fence shall be provided for the front return air opening.

3.4.5.2.2 Style A, class 3. The class 3 display cases shall be the same as the class 2 display case except that the class 3 display case shall have 2 additional shelves within the refrigerated area. The class 3 display case may be specified for all types.

3.4.5.2.3 Style A, class 4. The class 4 display case shall be designed for 80°F ambient (air conditioned) spaces, with a lower deck, and three additional shelves for merchandising products behind a refrigerated air curtain. The class 4 display cases may be specified for all types. The class 4 shall be for wall installation.

3.4.5.2.4 Style A, class 5. The class 5 display case shall be the same as the class 4 except the class 5 shall have 4 additional shelves within the refrigerated area.

3.4.5.2.5 Style A, class 6. The class 6 display case shall be the same as the class 5 except the class 6 shall have 5 additional shelves within the refrigerated area. The class 6 display case may be specified for all types, except type IV and V.

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3.4.5.3 Style B. The style B display case shall be a narrow center aisle single deep well deck and shall have adjustable racks for at least three positions to permit shallow displays of various depths. The style B display cases shall be for center aisle installation with a uniform top height front and back. The style B display cases may be specified only for types IV and V.

3.4.5.4 Style C. The style C display case shall be a double wide aisle type case with a divider separating the two compartments. Each compartment shall be equipped with adjustable racks capable of permitting shallow displays in at least three positions. The overall depth of the display case shall be 77 to 81 inches. The style C display cases shall be designed for center aisle installation with a uniform top height front and back. The style C cases may be specified only for types IV and V.

3.4.5.5 Style D. The style D display case shall be a wide island (two back-to-back cases) with adjustable racks to permit shallow displays in at least three positions. The overall depth of the display case shall be 82 to 88 inches. The style D display cases shall be for center aisle installation with a uniform top height front and back. The style D cases may be specified only for types IV and V.

3.4.5.6 Style E. The style E display case shall be a utility case for end display of meat or dairy products and shall have up to five adjustable shelves.

3.4.5.7 Style F. The style F display case shall be a single deck air curtain case designed to be installed into an existing cooler. The style F display case shall have three rows of adjustable shelves and rear loading doors or curtains. The curtains shall be of the same type and style used in the manufacturer's commercial products. Doors shall be hinged or sliding type. The hinged doors shall be insulated and the sliding doors noninsulated. An end kit shall be provided to effect a smooth blend of air curtain with existing cooler. The type and size of ends shall be as specified (see 6.2). When two or more style F display cases are to be joined a joint kit shall be specified (see 6.2). The style F display case shall have lights the entire length of the canopy. When specified (see 6.2), the bottom deck shall be equipped with two position, adjustable wire shelves.

3.4.5.8 Style G. The style G display case shall be a bottomless air curtain case designed for rear loading of roll-in racks. The style G display case shall be designed as type II case (see 1.2) with canvas sliding curtain enclosures. When specified (see 6.2), a removable shelf kit to be used in lieu of one or two of the roll-in racks shall be furnished. The size 2, style G display case shall be furnished with 2 roll-in racks and the size 3, style G display case shall be furnished with 3 roll-in racks. The roll-in racks shall conform to MIL-R-43979. Style G display cases shall be equipped with bumpers on the interior front and sides to protect against impact of roll-in racks. Bumpers are to be made of hard rubber material, and shall be capable of being removed and replaced.

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3.4.5.9 Style H. The style H end units shall be designed to provide an integral or wraparound appearance when joined to style C or D display cases. Materials, accessories, trim and finish shall match adjoining cases. Electrical, plumbing, waste, drain, refrigeration systems shall be in proportion to the adjacent cases. The overall length of the end cases shall be 77 to 81 inches when matching style C cases and shall be 82 to 88 inches when matching style D cases. The style H end units may be specified only for types IV and V cases.

3.4.6 Superstructures. Superstructure canopies shall be furnished with the display cases complete with light fixtures, trim and accessories, and shall be finished in colors for parts normally colored matching the display case on which it is to be mounted. Finish shall be baked enamel. Canopy superstructures shall be all steel or aluminum construction including back panel, shelves, reinforcing brackets, and supports; and shall be furnished as an assembled unit or as a prefabricated unit ready for assembly. Fluorescent light fixtures, with reflectors and bulbs, shall be installed the full length of the canopy top for the M-5 and M-6 superstructures. The M-7 superstructure shall have fluorescent lights with protective light shields installed full length on both sides of the truss under both top and bottom shelves. The height of the shelves shall be adjustable. The top shelf may be considered the canopy top. A separate switch for control of canopy lights shall be provided. High and low shelf canopies supplied with display cases having a rear sill with less than 8-1/2 inches front to back of level display surface, shall be furnished with an extra 12 inch wide shelf designed to be installed approximately 3 inches above the sill. Canopies and shelves shall be supported, reinforced, and constructed to support a uniform load of not less than 5 pounds per square foot of shelf without sagging or deforming below 1/4 inch (see 4.4.3), except that the loading does not apply to sliding glass or mirrored superstructure. Shelving for style M-7 shall be full length. End stops for 50 percent of shelf ends shall be furnished to close end of shelves to provide for shelf displays that are not continuous. Wire shelves shall be furnished for type I and flat metal shelves shall be furnished for other type cases.

3.4.6.1 Superstructures M-1, M-2, M-3, and M-4. Superstructures M-1, M-2, M-3, and M-4 shall extend the full length of the display case. Slanted mirrors shall be mounted at an angle to give full reflection of the display but shall not reflect any section of the floor from a straight ahead viewpoint. Slanted mirrors of high sliding type shall have an observation port 1 inch high across the width of each mirror. Sliding mirrors and clear glass shall be framed with stainless steel, chromium plated metal, aluminum alloy, and shall be mounted in the display case on tracks and bearings that permit easy operation. Frames shall have chromium plated metal, aluminum, or stainless steel handles. Sliding mirrors and clear glass doors shall be easily removable for cleaning and replacement. Tracks for doors shall have drain holes. Impact protection at top points of opening and closing shall be provided. Mirrors and windows shall be plate glass. Superstructures M-1, M-2, M-3, and M-4 are available only for style A, classes 1 and 2 in all types except type II.

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3.4.6.2 Superstructures M-5 and M-6. Superstructures M-5 and M-6 shall extend the length of the display case and shall be the dimensions specified. Intermediate shelves shall be 12 to 14 inches deep. A bottom shelf shall be furnished as a part of the shelf canopy and shall be as specified in 3.4.6. All intermediate and bottom shelves shall be furnished in multiples of 4 feet nominal length; shall be adjustable in height, 1-1/2 to 2 inches on centers; and shall be held solidly in place where installed. Shelves and support components including back plates shall be fabricated of the materials and finish specified in 3.4.6. Price tag mouldings to take 1-1/4 inch tags shall be installed for each shelf. Superstructure M-5 and M-6 are available only for style A, class 1 in all types of display cases, except type II.

3.4.6.3 Superstructure M-7. Superstructure M-7 for style C and D for types IV and V display cases shall consist of center supported truss type top and bottom open shelves extending full length of the display case with pilaster supports 4 feet nominal, on centers, or shall be of rigid construction designed to be supported by pilaster supports at each end. Shelf supports shall be braced and held rigidly in vertical position to insure stable shelving. Fluorescent lights with protective light shields shall be installed full length on both sides of under top and bottom shelves. The top shelf shall be 26 to 30 inches above top sill of display case, and bottom shelf shall be 10 to 14 inches below top shelf. Top shelf shall be 20 to 24 inches deep overall for style C cases, and 26 to 30 inches for style D cases. Bottom shelf shall be 12 to 14 inches deep overall for style C cases, and 20 to 22 inches deep for style D cases. Price tag mouldings to take 1-1/4 inch tags shall be installed full length on both sides of top and bottom shelves.

3.4.7 Framing. Framing members shall form a rigid structure capable of providing support during handling and shipping, and for the loaded shelves, racks, liner, shell, trim, and superstructure. Frames shall be reinforced and predrilled for bolting or locking cases together as required for lineup installations and for twin aisle installations. Reinforcement and backing of the outer shell and inner liner shall be provided as required for support and fastening of hardware and brackets. Reinforcement and backing of the shell and liner shall provide support to the drain and eliminate flexing of the outer shell, or inner liner during make-up of drain connections. Metal framing members shall be phosphatized steel finished with a rust and corrosion inhibitor, galvanized steel, aluminum, or corrosion-resisting steel. Reinforcing and framing members not totally inclosed, or within walls, shall be placed in a manner to be easily cleanable. The ends of all hollow sections of reinforcing and framing members shall be sealed. All vertical channel sections shall be either completely closed or open to the floor.

3.4.7.1 Support base. All display cases shall have rigid support bases fabricated from steel and recessed at the front of the case and shall be provided recessed splash-guards (kick-plate). Metal shall be the same type as specified for framing members in 3.4.7.

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3.4.8 Outer shell and inner liner. Access openings in the outer shell and inner liner shall be provided as required for assembly and installation. Clearance space and access openings shall be provided with fitted covers and shall be thoroughly sealed after fabrication to prevent moisture or water vapor from entering the insulated space. Interior walls may have openings to permit vapor transmission from the insulation, provided such openings are located and protected to prevent the entry of free water from any source. All space between the outer shell and inner liner shall be filled with insulation of the types specified in 3.3.1.1, 3.3.1.2, or 3.3.1.3. Rigid or foamed-in-place insulation shall be used in the bottom space to support and to prevent any displacement of the liner, except that when low conductive supports are installed, semirigid insulation may be used.

3.4.8.1 Outer shell. The front, top, and ends of the outer shell shall be enameling iron, aluminum, or carbon steel. The bottom shall be stainless steel, aluminum, or galvanized steel. The rear shall be enameling iron, aluminum, carbon steel, or galvanized steel.

3.4.8.2 Inner liner. The inner liner or the insulated panels shall be aluminum alloy, fiberglass, or galvanized steel. When galvanized steel is used, the inner side shall not be exposed and shall be protected and concealed by the inside air baffles. Air baffles shall be plastic, aluminum, or steel finished with baked enamel, or shall be aluminum alloy except that inside front baffles (excluding style C and D) and lower deck baffles may be minimum spangled galvanized steel. Inside ends shall be of the materials specified for air baffles or as specified in 3.5.2. Seams shall be smooth so that interior of cabinet can be kept clean and sanitary. Tapping plates and structual reinforcement, as required, shall be secured to the insulation side of the inner liner to provide support for brackets, shelves, baffles, or other items to be fastened to, supported by, or hung on the exposed side of the inner layer. Duct openings in the display area shall be protected from accidental entrance of displayed items or foreign matter.

3.4.9 Drain. A drain pipe no smaller than 1 inch iron pipe size (I.P.S.) shall extend through the bottom of the display case. Type I, class 1 or 2 produce display cases shall have a minimum 2 inch waste drain; type II, styles B, C, and D shall have a minimum 1-1/2 inch drain. The drain hole and drain pipe shall be accessible for normal installation and cleaning without removal of any major components. The display case drain pan beneath the coils and the case bottom shall provide drainage so that free water does not stand on the drainage surfaces when cases are level. Drains shall be completely sealed to the case to prevent moisture from entering insulated space. All condensation shall be drained without dripping into storage areas. A length of 1 inch pipe, with formed trap located at least 18 inches from the drain nipple, shall be furnished with each type case. When a drain trap is built into the cases, the external trap is not required. The drain trap shall be brass, aluminum or PVC. Pipe and fittings shall be brass, aluminum, galvanized iron or PVC.

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3.4.10 Cart bumper. Display cases shall be provided with a cart bumper of stainless steel, aluminum, or high-impact-resistant styrene plastic. Cart bumpers may be used as a raceway for electrical conductors.

3.4.10.1 Corner guards. When corner guards are specified (see 6.2), they shall be constructed of 10 gauge (.1350 inch), type 304 stainless steel, electra polished, with flange to be lagged to the floor. The corner guard shall be 6 by 6 by 24 inches high.

3.4.11 Glass front. The front glass, when provided (see 3.5.1) for cases, shall rise 2-1/2 inches to 10 inches above the top of the front sill. The glass used shall be 1/4 inch thick safety or tempered glass or shall be 2- or 3-glasspane sealed units, as required, to eliminate fogging. Glass fronts shall be capped with a channel section and supported vertically. All exposed channel section, supports, and trim for glass fronts shall be chromium plated brass, extruded and polished aluminum, or stainless steel. Glass fronts shall not project above case ends at their junction.

3.4.12 Hose and spray nozzle. Type I produce cases shall be equipped in the front part of the case with a water hose and a chromium plated brass nozzle to provide a fine mist spray. Hose shall be of sufficient length to service entire display area and shall be automatically retractable to a retainer in a concealed position. Copper water supply service connection shall be extended from hose to exterior of case and terminated with a brass shut-off gate valve located in a recess within the outline of the display case.

3.4.13 Bottom storage (double duty cases). The bottom storage area for type III cases shall be constructed with a center shelf and with floor racks. Shelf and racks for storage area shall be steel, hot tin dipped or copper-nickel-chromium finished, anodized aluminum, or baked enamel on electro zinclated or phosphatized steel.

3.4.14 Platters. Unless otherwise specified (see 6.2), meat platters, approximately 10 inches by 30 inches, in sufficient quantity to cover the shelf and floor rack of the storage area in the type III double-duty cases shall be furnished with each type III double-duty display case. Unless otherwise specified (see 6.2), the platters shall be anodized aluminum or stainless steel.

3.4.15 Shelves for display case. The shelving located in the upper part of class 2, 3, 4, 5 and 6 display cases shall be baked enamel on steel or aluminum, stainless steel or aluminum. Color of applied finishes shall be white. Shelves and supports shall be designed to support the loading specified for super-structure shelves (see 3.4.6 and 4.4.4). All shelves shall have a full length tag moulding to take 1-1/4 inch tags located on the front of the shelves. Depth of shelves for style A, class 2 and 5 for type I cases shall be 18 to 20 inches; shelves for type II, 20 to 22 inches; type III, 19 to 22 inches; type IV and V, 18 to 23 inches.

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3.4.15.1 Peg bars and hooks. When specified (see 6.2), stainless steel peg bars and peg hooks shall be furnished (in lieu of upper shelves) for hanging displays. Peg bars shall be 48 inches long. Peg hooks shall be 8 to 13 inches in length.

3.4.16 Deck racks. Deck racks for display cases shall be baked enamel on electro-zinc plated or phosphatized steel, stainless steel, anodized aluminum alloy, aluminum, hot tin dipped steel, or copper-nickel-chromium finished steel.

3.4.17 Evaporators. The evaporators shall be constructed of copper or aluminum tube with either copper or aluminum fins. Fins shall be bonded to tube to assure a high conductivity. Evaporator shall be dehydrated and sealed. Refrigerant connections shall be easily accessible for servicing.

3.4.17.1 Fans. Fans shall provide positive forced circulation of the refrigerated air to maintain uniform temperature throughout entire display area to be refrigerated. The fan and hub shall be made of electro-tin plated brass, aluminum alloy, or stainless steel. The fan shall be connected directly to the motor. The motor and fan shall be accessible for servicing and replacement. When specified (see 6.2), a fan kit shall be provided for use at the base for back to back placements of display cases.

3.4.18 Condensing units. The display cases shall have air cooled, remote or self-contained condensing units, or water cooled, remote or self-contained condensing units, as specified (see 6.2). The condensing unit shall consist of a motor, compressor, condenser, receiver, motor safety devices and other necessary equipment all mounted and arranged on a unit base or rack. The unit base or rack shall be shock mounted to reduce vibration to a minimum. Unless otherwise specified (see 6.2), condensing units for self-contained cases designed for 50- or 60-hertz operation, and remote units designed for 60-hertz operation, shall be semi-hermetic or hermetic, either thermally or mechanically sealed. Remote units designed for 50-hertz operation may be open type or hermetic, thermally or mechanically sealed. The condensing units procured for operations on a frequency of 50-hertz shall operate with full load for 4 hours at the specified frequency and at 90 percent of the specified system voltage followed by operation for 4 hours at specified frequency and at 110 percent of specified voltage, and shall not cycle on the motor protector (see 4.4.1.2c). All condensing units shall be piped, valved, wired, and serviced ready for operation and connection to utility sources as required. The assembled unit shall be dehydrated and sealed; remote units shall contain a holding charge, and self-contained units a full operating charge of the recommended refrigerant oil. The compressor shall contain the proper amount of manufacturer's recommended oil. When multiple display case lineups are to be operated by one condensing unit, not over 48 feet of display case shall be operated by one condensing unit. The refrigerant shall not contain moisture in excess of 10 parts per million (P.P.M.) (see 4.4.4).

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3.4.18.1 Condensing unit base and enclosure. Condensing units for self-contained cases shall be mounted on a unit base and for remote cases shall be mounted on a unit base or rack. The unit base and rack shall be constructed of structural steel shapes welded to form a rigid structure. The condensing units, control panel, and accessories complete and ready for connection as specified in 3.4.18 shall be mounted on the unit base or rack. When specified (see 6.2), a prefabricated watertight, heavy gauge steel, aluminum housing shall be provided for outdoor weather protection of condensing units. The outdoor protective housing shall be constructed with steel frame, hinged panels on front, single stand, crankcase heater, condenser screen, and louvered exhaust. The housing shall be constructed as a removable enclosure so condensing units may be exposed for maintenance and service.

3.4.19 Controls and accessories. All controls and accessories required for satisfactory operation of the display case, when connected to the condensing unit, shall be supplied and shall be piped and wired and ready for use. The refrigeration system shall include adjustable thermostatic expansion valves, heat exchanger as required, dehydrator, dual pressure control with adjustable differential, compressor service valves, oil separator if necessary, liquid line receiver shut-off valve, combination moisture and liquid indicator, temperature alarm system, and control devices such as demand defrost controls and dewpoint controllers as necessary for effective automatic operation and defrosting. A time-pressure or time-temperature clock electric defrosting system for time initiated and pressure or temperature terminated defrost shall be provided. Refrigeration shall be restarted automatically by pressure or temperature. The clock shall include a safety device which shall automatically terminate the defrost period by time if the pressure or temperature fails to reach the desired cut-in point. A dial type thermometer shall be mounted to show the temperature of the air leaving the display area and positioned to be clearly visible from the center front of the display case. Case temperature shall be controlled by pressure settings at the condensing unit. Any transformers required for operation of the control circuits or lighting and cooling fan motors on 120-volt electrical current shall be furnished with the units supplied. When specified (see 6.2), a sensor probe (for connection to the store temperature sensor alarm system) shall be installed in the air discharge area.

3.4.20 Motors. Unless otherwise specified (see 6.2), motors rated under 3/4 horsepower shall be designed for operation on nominal 120-volt, 1-phase, 60 hertz alternating current electrical system, and motors rated 3/4 horsepower or over shall be designed for operation on nominal 240-volt, 3-phase, 60 hertz or 208-volt, 3-phase, 60 hertz alternating electrical system. All display cases shall be completely electrically wired and shall conform to NFPA 70. Compressor motors shall have sufficient capacity to start the refrigerating system under full load and to operate in high ambients specified without cutting out on overload. When frequency other than 60 hertz is specified, the motor shall be specifically designed for the frequency cited and the frequency shall be included with the data on the nameplate. Unless otherwise specified (see 6.2), evaporator for fan motors, controls, and lights shall be designed to operate on

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120-volts, 1-phase, 60 hertz. When specified for European use, compressors shall be designed to operate on 380-volt, 3-phase 50 hertz, and evaporator fan motors, controls and lights shall be designed to operate on 220-volts, 1-phase, 50 hertz. An integral transformer shall be supplied to step down the voltage for control system operation.

3.4.20.1 Motor starter. Automatic starters shall be supplied for all 3-phase motors and for all 1-phase motors over 1 horsepower. The compressor motor starter shall be across-the-line type, continuous duty, suitable for automatically starting the condensing unit compressor motor when actuated by the automatic controls. The starter shall be mounted in a drip-proof enclosure. The motor starter shall be equipped with a thermal overload protector if the motor is not otherwise equipped with such protection. All relays shall be the "quick-trip" type, factory calibrated for the particular motor application.

3.5 Finish.

3.5.1 Display case exterior. The exterior of the display case shall be finished in baked synthetic enamel, as standard with the manufacturer and shown in current published catalogs and sales literature. Unless otherwise specified (see 6.2), the exposed top, fronts, back and ends shall be white or color conforming to FED-STD-595 as specified (see 6.2). The bottom of cabinets, and the backs of units specified for wall installation shall be aluminum or one of the materials specified in 3.4.8.1 unpainted or a black or white color. The kick-plate or splash guard shall be black or in a color and finish to match the trim. Trim shall be manufacturer's standard in black baked porcelain enamel, stainless steel, or aluminum alloy finish standard with the manufacturer.

3.5.2 Display case interior. The interior of the display case shall be stainless steel or aluminum alloy finish with the manufacturer; baked enamel on phosphatized steel; or any combination of the foregoing unless otherwise qualified by these specifications. The interior front and bottom may be unpainted minimum spangled galvanized steel. Manufacturer's recommended load line shall be clearly marked and identified in a contrasting color. In double duty cases, the interior liner of the storage area shall be the finish standard with the manufacturer. Plastic product dividers shall be provided for type I display cases on the basis of one for each 2 linear feet.

3.5.3 Superstructures. All components of the superstructures including ends, backs, tops, pilasters, structural supports, shelves, braces, dividers, and accessories shall be finished in baked enamel and fitted with trim to match the display case.

3.5.4 Application.

3.5.4.1 Baked enamel. Baked enamel finishes shall be a synthetic baked enamel applied to surfaces that have been cleaned and primed. The finish shall be smooth, applied uniformly, durable, tightly adhering, and nonabsorbent.

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There shall be no visible corrosion when the finished surfaces are wiped with a damp cloth after being exposed to a salt spray for 96 hours (see 4.4.5).

3.6 Data-name plate. The display cases and condensing units shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required and (D) Optional marking information shall not apply. The data-name plates shall be made of minimum 20 gage (0.035 inch) corrosion-resisting metal and attached to the display case or condensing unit by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved or applied by photosensitive means:

National Stock Number
Procurement Contract Number
Specification Data
Manufacturer's Name, Address and Telephone Number
Supplier's Name, Address and Telephone Number (List only if different from Manufacturer.)
Manufacturer's Model Number
Government Approved Manual Number (when specified, see 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the display case or condensing unit.

3.7 Instruction manual. An instruction manual shall be included with each complete assembly delivered. The manual shall include electrical wiring diagrams, refrigeration system diagrams, operation instructions, assembly instructions, parts lists and nomenclature.

3.8 Workmanship. All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. All sharp corners, edges, or projections on the interior or exterior shall be eliminated. Sheet metal or structural sections used in fabrication of the units shall be free of kinks, bends and wrinkles. Shearing and trimming shall be done accurately and cleanly on all portions of the work and shall be neatly finished. All bolt and rivet holes shall be accurately and cleanly punched or drilled. Rivets, when used, shall be driven to completely fill the holes, with full-sized heads neatly finished and in full contact with the surface of the member. Bolts, nuts, and screws shall be drawn tight and so secured that they will not work loose under vibration. Optional equipment, accessories, and piping shall be securely fastened and braced to prevent damage or loosening during transit or operation.

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3.9 Servicing after testing. Upon completion of testing, the display case shall be allowed to come to equilibrium at room temperature. The interior and exterior shall be dried, cleaned, inspected, and required adjustments, including setting temperature controls for specified operating conditions, shall be made prior to final preparation for packing and packaging. Remote condensing units shall be disconnected, piping sealed, and the condensing units prepared as specified in 3.4.18.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of reference documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.2.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.2.2.1 Examination of operations . Examination shall be made at any point or during any phase of the manufacturing process to determine that the required operations or assemblies are accomplished as specified. Whenever nonconformance is noted, corrections shall be made on affected items and process.

(a) Determine reinforcement and backing of shell and liners is provided as well as predrilled, (see 3.4.7).
(b) Determine that fitted covers of the access openings are in place (see 3.4.8).
(c) Determine that insulation is installed as specified in 3.4.8.
(d) Attachment of backing plates, tapping plates, and reinforcing to determine compliance with 3.4.8.2.
(e) Application of baked enamel finish (see 3.5.4.1).
(f) Punching and drilling of bolt and rivet holes (see 3.8).
(g) Piping is fastened or braced (see 3.8).
(h) All metal forming is as specified (see 3.8).
(i) Shearing and trimming accurate and clean (see 3.8).

4.2.3 In-process tests. Any in-process testing failure shall be cause for rejection of all components represented by the applicable sample.

4.2.3.1 Organic enamel test. The salt spray test as specified in 4.4.5 shall be performed (on test specimens prepared at the time) the initial production unit is fabricated. The size of the sample specimens shall be 3 inches by 6 inches. As a minimum, one representative sample of each component being finished with baked enamel shall be tested.

4.2.3.2 Testing of density of insulation. The density test for insulation shall be performed as specified in 4.4.6. The inspection lot shall be the amount of material used in one day's production. The sample size for this test shall be a 6 inch cube of insulation.

4.2.4 End item inspection. The sample unit for this examination shall be one completely fabricated display case. The inspection lot shall consist of all display cases of one type, style, class, size, ambient temperature operation, and superstructure offered for inspection at one time.

4.2.5 End item visual examination. The end items shall be examined for the defects listed in table IV. The lot size shall be expressed in units of display cases. The sample unit shall be one display case. The inspection level shall be level II (see 6.4).

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TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not color specified	101	
	Not finish specified	102	
	Load line not as specified	103	
Design, construction and workmanship (applicable to all components and assemblies)	Part missing, not in proper alignment, or not type specified	104	
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform function intended (condition cannot be readily rectified)	105	
	Adjustable assembly not properly adjusted but can be adjusted		201
	Component not attached as specified	106	
	Component not accessible for servicing where required		202
	Access openings not as specified	107	
	Sharp edges, burrs or slivers not removed	108	
	Bolts, nuts, screws, studs, or other type of threaded fasteners, missing, broken, stripped, fractured or loose		203
	Sheet metal has kinks, bends or wrinkles		204
Rivets (where applicable)	Rivets not completely driven, heads not full size, not in contact with both surfaces		205
Wiring and electrical	Wiring not as specified	109	
	Wiring diagram missing, incomplete, not legible, not affixed in prescribed manner		206
	Lighting not as specvified	110	
Data name plate	Omitted or not as specified		207
	Information incomplete or illegible		208
	Not readily visible to the operator		209

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4.2.6 End item dimensional examination. The end item shall be examined to determine compliance with the dimensions specified in section 3. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of display cases. The sample unit shall be one display case. The inspection level shall be level II (see 6.4).

4.2.7 End item testing. Each production unit shall be tested as specified in 4.4.4. Any nonconformance shall be cause for rejection of the display case. One display case from the lot shall be tested as specified in 4.4.1 through 4.4.3. Any nonconformance shall be cause for rejection of the inspection lot.

4.2.8 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.4).

Examine	Defect
Marking (exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing Floating mechanism not secured All components and packages not positioned or secured as specified
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container

4.3 Code and standard compliance. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.2, shall be examined and validated as proof of compliance. Prior to Government approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the display cases he proposes to furnish under this specification meet the applicable requirements of UL as follows:

4.3.1 Underwriter's Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriter's Laboratories standards specified in 3.2 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the display cases have been tested and conform to the applicable requirements of the specified UL standard.

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4.4 Methods of inspection. Prior to approval of the first shipment, the contractors may submit, in lieu of the testing requirements specified in 4.4.1 through 4.4.6, certified test data and test results for each type, style, and size display case to be delivered under the contract. The test results shall show, to the satisfaction of the contracting officer, that the display cases will conform to the test requirements of this specification. In addition, if the contract requires delivery of a display case assembly consisting of two or more of the same type and style display cases, and the assembly is to be operated by one or more condensing units, the contractor shall certify that the capacity of the condensing unit(s) to be furnished with each assembly (line-up) shall equal or exceed the combined refrigeration capacity requirements of all display cases in the assembly (line-up).

4.4.1 Performance test.

4.4.1.1 Display case preparation.

(a) The display case with the refrigeration system shall be assembled and set up for the performance test of 4.4.1.2. All shelves, racks, plugs, and partitions shall be in place in accordance with the requirements specified.

(b) The refrigeration system shall be charged; the refrigeration unit shall be given a run-in operating period of approximately 8 hours at room temperature; controls shall be checked; and the power supply voltage adjusted to ±5 percent of the voltage specified for design and operation, and other electrical characteristics shall be as specified.

(c) Placement of the display case in the test room shall be made to simulate conditions in an installed location, that is, wall cases shall be backed up to a wall space, the ends closed off with a temporary partition, and at least 5 feet of clear space provided at the front of the case. Center aisle cases shall have at least 5 feet of clear space all around the display case.

(D) Air circulation in the room shall be such that the specified uniformity of temperature in the heated room is obtained. The display case shall be shielded from forced air currents. Ambient temperatures shall be measured at points 1 foot above the floor and 3 feet above the cabinet at the center point on each side. Ambient temperatures at each point shall be maintained within ±2°F of the specified test conditions. The test conditions shall be maintained during stabilization periods as well as during test runs. Where cases are adjacent to walls or partitions, the temperature of such walls and partitions shall be the same as the room temperature.

(e) In preparation for any operating test, the display case shall be operated under specified test conditions for sufficient time to establish steady or repetitive state. Immediately after this preparatory period, the display

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case shall be operated for the specified test period during which the specified elements of performance are determined and recorded. Temperature observations shall be made continuously or at equal intervals of not greater than 15 minutes. Reported temperatures shall be the arithmetical average of all readings taken at equal time intervals.

(f) The display case (display area and storage area) shall be loaded with simulated load corresponding to the manufacturer's recommended maximum load.

(g) The interior display case temperature shall be measured with not less than eight thermocouples or other temperature measuring devices, located in the top of the interior load and equally distributed across the top of the display compartment. Double-duty cases shall have four additional thermocouples equally distributed throughout the load in the bottom storage compartment.

(h) The condensing unit of remote cases shall be located at least 5 feet from the display case with the refrigerant lines uninsulated and specified ambient air entering the condenser. Condenser inlet air temperature of both remote and self-contained condensing units shall be determined by averaging the readings of four temperature measuring devices located 1 to 2 inches from the condenser fins and equally distributed over the inlet air area.

(i) The electric defrosting device shall be made inoperative for the performance test.

(j) Prior to the start of the performance test, the display cases may be precooled to the temperatures specified below:

Type	Precooled temperature
I	32°F
II	31°F
III	28°F
IV	-10°F to -20°F
V	-18°F to -25°F

(k) Where thermometers are used to obtain temperature data for performance requirements, they shall be numbered and each one shall be located in a fixed position for the duration of the test, and shall not be moved except as required to take readings at the fixed position. Thermometer readings for either bulb or glass type, shall be made to an accuracy of 1/2 a thermometer graduation.

4.4.1.2 Procedure.

(a) The condensing unit of the display case shall be placed in operation when the condensing unit ambient temperature is 100°F for remote units. The display case ambient temperature, determined as the average of the readings of the temperature measuring devices, is 78°F and 55 percent relative humidity or

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90°F, as specified. During the test period, observation shall be made that the interior display case temperature is as specified in table III, determined as the average of the temperature measuring devices located in the display case. An observation shall be made that no single interior temperature (regardless of average) and none of the running times shall exceed the interior temperature and running time requirements specified in 3.4.4.2. Types IV and V self-contained cases may be tested with night covers specified in 3.4.4.2 in place for the 90°F ambient requirement.

(b) Upon satisfactory completion of the controlled temperature display case performance tests, the remote condensing unit ambient temperature shall be raised to 110°F while the display case ambient temperature is held constant. Ambient temperatures for self-contained display case tests shall be raised to 100°F. The display cases shall be observed for a period of 4 hours at these elevated temperatures without cutting out on overload to conform to the overload requirements of 3.4.4.2. During operation in the elevated ambient temperatures, the contractor may return the electric defrost devices to operating condition, and the internal display case increased temperatures shall be taken every 30 minutes and made a matter of record in the test data compilation.

(c) Refrigeration units procured for operation on a 50 Hz frequency shall be tested for conformance with 3.4.18. Ambient display case conditions for this test shall be normal room temperature, 60°F to 100°F. The motor shall not be damaged and shall not cut out on either inherent or external overload protectors during these tests.

4.4.2 Exterior temperature test. To determine exterior surface temperature conformance to 3.4.4.2, the display case shall be operated for a period of not less than 4 hours under the conditions specified in 4.4.1.2. During and at the end of the 4 hour test the entire exterior of the display case shall be inspected in 14 foot quadrants to determine that exterior surfaces are not more than 10°F below room ambient test temperatures.

4.4.3 Shelf load tests. Canopies and shelves for the display case and superstructures shall be assembled and mounted as intended for use in their service installation. To determine uniform load compliance with 3.4.6 and 3.4.15, a load of 50 pounds per square foot shall be distributed uniformly on the entire top area of all canopies and on all shelf areas. Material for loading shall be nonrigid and shall be in segments with a base area of not more than 72 square inches. When fully loaded it shall be observed that the canopies and shelves shall show not more than 1/4 inch deflection from the horizontal and user removal of load, the supports, canopies and shelves shall show no deformation or permanent set.

4.4.4 Operation test. Display cases and condensing units shall be operationally tested, electronically tested, to determine compliance with operation requirements of 3.4.4.3. Display cases for remote condensing units need not be connected to a condensing unit nor perform any refrigeration function during

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this test. During and after this test, for self contained cases, the combination moisture and liquid indicator shall be observed and shall indicate "dry" within the limits specified in 3.4.18. A liquid line temperature of not more than 100°F shall be maintained when observing the indicator. If the indicator shows that there is moisture in the system, drier replacements shall be made and the system operated until the moisture indicator shows dry.

4.4.5 Salt spray test. Metal test specimens, coated with organic material shall be subjected to salt spray test in accordance with ASTM B 117 for 96 hours. The test specimens shall be representative of components of the display case. After exposure to the salt spray for 96 hours, the surface of the specimens shall be wiped with a damp rag and an examination for corrosion made to determine compliance with 3.5.4.1.

4.4.6 Density test for insulation. When rigid or pre-expanded insulation is used a 6 inch cube shall be cut and weighed. This weight, 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance to the density requirements of 3.3.1.1. When foamed-in-place insulation is used, one free foam sample of the insulation specified in 3.3.1.2, a minimum size 8 inch cube, shall be taken prior to each day's production. From the center of the sample, a 6 inch cube shall be cut and weighed. This weight, 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance with the density requirement of 3.3.1.2. In addition, a daily record of the foam insulation shall be kept as follows and the production unit shall not deviate more than ± 5 percent from the data recorded on the unit used for the test specified in 4.4.2.

Calibration of mix (mix proportions)
Amount of flow (per unit)
Time of flow (per unit)
Mould temperature (per unit)

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Preservative. The refrigeration system shall be prepared for shipment in accordance with the manufacturer's recommended procedure for overseas shipment or as otherwise specified. Drive belts shall be removed, and exposed uncoated ferrous metal surfaces of pulleys and shafts shall be cleaned in accordance with process C-1 of MIL-P-116 and coated with primer conforming to TT-P-664. When the primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed. All floating mechanisms and tubing not rigidly fixed, such as vibration loops, free coils, capillary tube loops and coils shall be secured in a fixed position to minimize vibration and subsequent damage during handling and shipment. Floating mechanism is defined as any part that is rubber, spring, bracket, or otherwise mounted which might vibrate during shipment.

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5.1.1.2 Unit packing.

5.1.1.2.1 Removable components.

5.1.1.2.1.1 Mirrors, glass and glazed doors. When removable mirrors, glass and glazed doors are supplied, they shall be removed and individually unit packed in corrugated fiberboard boxes conforming to style FOL, type CF, variety SW, class domestic of PPP-B-636. The opening shall be on the panel with the longest dimension and the adjacent shortest dimension. A minimum of 1-1/2 inch clearance shall be maintained between the sides of the glass or mirror and the interior surfaces of the fiberboard box by means of spring pads made of the same material as the box, placed at each edge of the mirror, glass, or glazed door. Box closure shall be in accordance with the appendix of PPP-B-636. The boxes containing the mirrors, glass, or glazed doors shall be positioned on edge (vertical position) and shall be blocked or braced within the well of the display case or inside the shipping container in a manner to prevent shifting during shipment and handling.

5.1.1.2.1.2 Shelves, racks, platters, or trays. The removable shelves, racks, platters or trays shall be unit packed in a corrugated fiberboard box or boxes conforming to style RSC or OPF, type CF, variety SW, class domestic of PPP-B-636. Box closure shall be in accordance with the appendix of PPP-B-636. The fiberboard box or boxes shall be placed within the well of the display case and shall be immobilized in a manner to prevent damage or shifting during shipment and handling.

5.1.1.2.1.3 Superstructures. Superstructures which are normally demounted for shipment shall be placed in the well of the display case. The end mounting bolts of the display case shall be fitted with spacers to permit free entry of the superstructure within the interior of the case. The superstructure shall be immobilized in the well of the display case in a manner to prevent shifting during shipment and handling. All blocking and bracing in direct contact with the display case or superstructure shall be cushioned with cellulose wadding or fiberboard to prevent abrasion of enameled or porcelainized surfaces.

5.1.1.2.2 Fixed components and components not normally removed for shipment.

5.1.1.2.2.1 Fluorescent bulbs. All installed fluorescent bulbs shall be secured in place with pressure-sensitive tape.

5.1.1.2.2.2 Glass, mirrors, and glazed doors. All fixed or nonremovable glass, mirrors, and glazed doors shall be protected in a manner to prevent breakage during shipment and handling.

5.1.1.2.2.3 Shelves and racks. Shelves and racks normally shipped in place shall be secured in place in a manner to prevent damage in shipment and handling.

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5.1.1.2.3 Miscellaneous parts, instruction manual, and assembly hardware. All miscellaneous component parts and assembly hardware shall be unit packed method III of MIL-P-116. The packaged components and assembly hardware shall be secured in place within the well of the display case in a manner to prevent damage or shifting during the shipment and handling.

5.1.2 Commercial preservation. Each display case shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Display case. Each display case, preserved as specified in 5.1, shall be packed in a sheathed, nailed wood crate conforming to type I, class 1 or 2, style a or of MIL-C-104. The boxes containing doors, not positioned within the well of the display case shall be positioned on edge (vertical position) and secured within an unused portion of the shipping container. Anchoring and cushioning of contents shall be in accordance with the appendix of MIL-C-104.

5.2.1.2 Condensing units. Each remote condensing unit shall be packed in a nailed wood shipping container conforming to class 2, style 2 or 4, grade A or B, type 3 load of PPP-B-621. Units shall be secured within the shipping container with standard carriage or step bolts. The size of the bolt shall be determined by the diameter of the hole in the base of the item. A lockwasher shall be used under the nut. Bolts securing the unit in the shipping container shall pass through the skids; bolting through the base only is prohibited. Bolt heads shall be countersunk into the side runners and no portion of the bolt head shall extend below the bottom surface of the skid runner. A minimum of one inch clearance shall be provided between the unit and the inside surfaces of the side, end, and top panels of the shipping container. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. All seams of the shroud shall be sealed with a 3/4 inch minimum width continuous seam of water-resistant adhesive conforming to MMM-A-260. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-621.

5.2.2 Commercial packing. Each display case, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The display cases covered by this specification are intended for use in establishments dispensing food, for the refrigerated display of products.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification
b. Type, style, class, size, suffix designation, superstructure, and ambient temperature operation of display case (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When glass other than tempered glass or safety glass is required (see 3.4.1)
e. When 3 inch price tag moulding is required (see 3.4.2)
f. Information required for matching lineup (see 3.4.3)
g. When 78°F or 90°F ambient operating temperature is required (see 3.4.4.2)
h. When access doors to storage space shall be in the front or rear of display cases (see 3.4.5.1)
i. Type and size of end required for style F display case (see 3.4.5.7)
j. When adjustable wire shelves are required (see 3.4.5,7)
k. When a joint kit is required to join two or more style F display cases (see 3.4.5.7)
l. When a removable shelf kit is required (in lieu of roll-in racks) (see 3.4.5.8)
m. When corner guards are required (see 3.4.10.1)
n. Size of platters if other than specified (see 3.4.14)
o. Whether material for platters is other than specified (see 3.4.14)
p. When peg bar and peg hooks are required (see 3.4.15.1)
q. When a fan kit is required (see 3.4.17.1)
r. Whether condensing units shall be air cooled, water cooled, remote or self contained (see 3.4.18)
s. When other than hermetic type condensing units are required (see 3.4.18)
t. Whether condensing unit enclosure is required (see 3.4.18.1).
u. When sensor probes for central alarm systems are required (see 3.4.19)
v. Electrical characteristics if other than specified (see 3.4.20)
w. Color, if other than white (see 3.5.1)
x. Government approved manual number to be included on the data-name plate when manuals are purchased under the contract (see 3.6)
y. Levels of preservation and packing (see 5.1 and 5.2)
z. Acceptance criteria required (see 6.4)

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6.3 Definitions. The following definitions are applicable to this specification:

Accessible: Accessible shall mean readily exposed for proper and thorough cleaning and inspection with the use of only simple tools, such as a screwdriver, pliers, or open-end wrench.

Cleaning: The term cleaning shall mean the physical removal of residue of dirt, dust, foreign material, or soiling ingredients or materials.

Readily cleanable: Readily (or easily) cleanable shall mean readily accessible and of such material, finish, and fabrication that soil may be effectively removed by normal cleaning methods.

Readily removable: Readily (or easily) removable shall mean capable of being taken away from the main unit without the use of tools.

Smooth: The word "smooth" is used to define a surface free of pits and inclusions and a cleanability equal to commercial grade hot rolled steel free of visible scale.

Sealed: Spaces required to be "sealed" shall have no openings that shall permit the entry of insects, rodents, dirt, or moisture seepage.

6.4 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in acceptance with MIL-STD-105.

6.4.1 For end item visual examination. An acceptance quality level (AQL), expressed in terms of defects per hundred units, of 2.5 for major defects and 6.5 for minor defects is recommended.

6.4.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.4.3 For packaging examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.5 Subject item (key word) listing.

Dairy products
Frozen food
Ice cream
Meat
Produce

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL

Review activities:
Army - MD, ME
DLA - GS

Preparing activity:
Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-0040605D(GL)	Display Cases, Mechanically Refrigerated, Self- Service

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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